

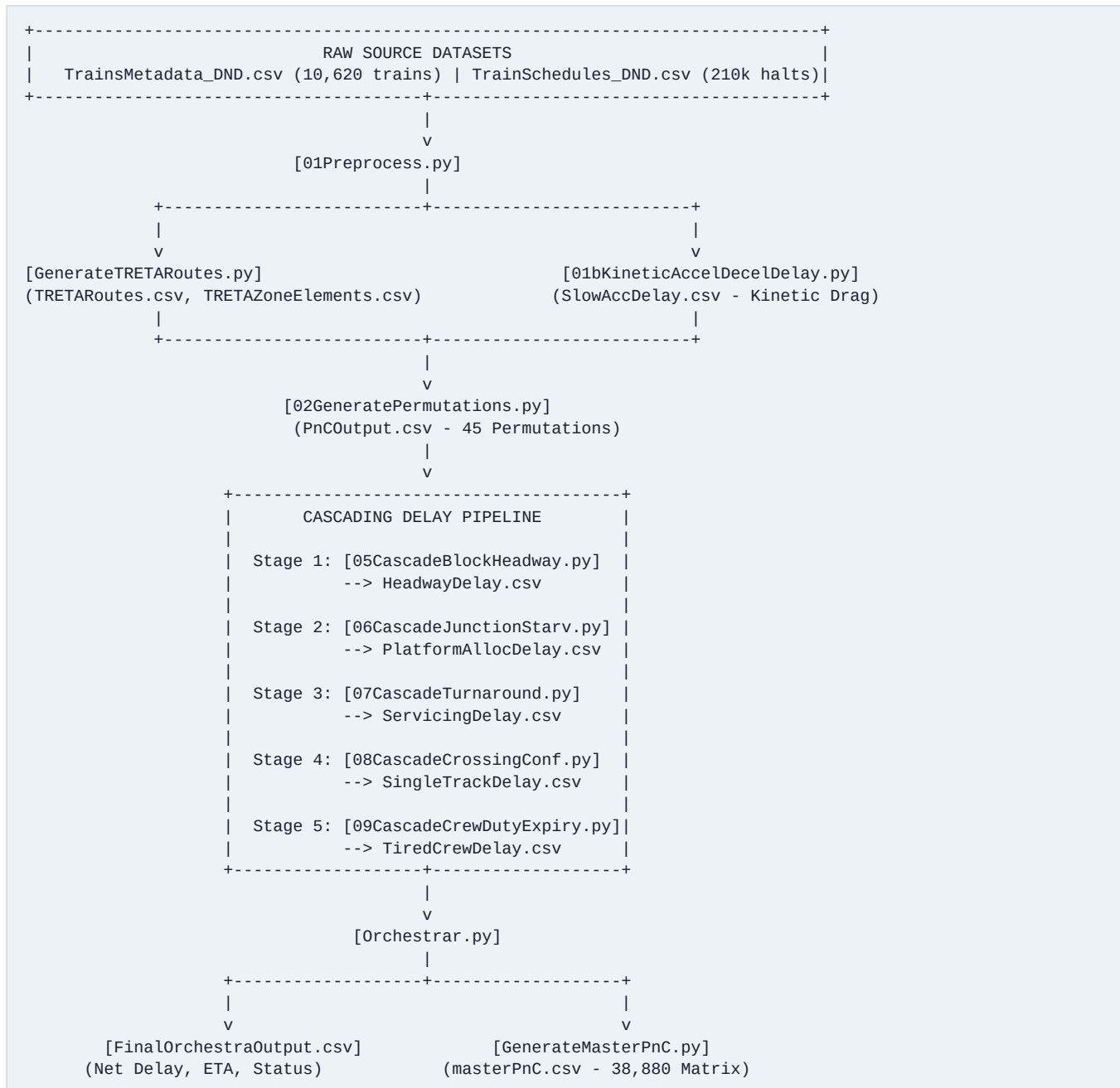
Indian Railways Delay Pipeline: Complete Architectural Specification

This document provides a comprehensive technical breakdown of every script in the [BlackSheep/Pipelines/](#) directory. It details the operational purpose of each module, the input files referenced, the output files created, and every attribute and metric calculated throughout the multi-stage simulation pipeline.

1. Executive Pipeline Architecture Overview

The Indian Railways Cascade Simulation Engine operates on a **Three-Tier Computational Architecture**:

- 1. Primary Shock Layer:** Models localized, exogenous physical disturbances (weather penalties, track speed restrictions / TSR, locomotive tractive effort, and kinetic deceleration/acceleration braking cycles).
- 2. Secondary Cascading Propagation Layer:** Propagates knock-on network ripple delays through 5 sequential bottleneck stages (block headway, platform starvation, rake turnaround, single-track crossing conflicts, and crew duty timeout).
- 3. Timetable Slack (EA) Absorption & Net ETA Layer:** Balances gross accumulated disturbances against allotted timetable cushioning ([extra_time_mins](#) and [net_slack_mins](#)) using tier-calibrated recovery factors to determine final punctual arrival ([ON_TIME](#) vs [LATE](#)).



2. Exhaustive Per-File Analysis

2.1. 01Preprocess.py

- **File Name:** `BlackSheep/Pipelines/01Preprocess.py`
- **Exact Role & Purpose:**

The foundational data engineering script for the entire codebase. It parses raw, non-standardized NTES timetable strings, merges schedule halt records with fleet metadata, assigns operational priority tiers (**T1_PREMIUM**, **T2_SUPERFAST**, **T3_EXPRESS**), and calculates the physical running parameters, scheduled halt durations, kinetic braking allowances, and the true calibrated Extra Time / Net Slack (**EA**).

- **Referenced Files (Inputs):**

- **Train-Data/TrainsMetadata_DND.csv**: Raw NTES train metadata for 10,620 trains (travel time, train types, source/destination).
- **Train-Data/TrainSchedules_DND.csv**: Raw timetable halt sequences and distances across 210,746 station stops.
- **Created Files (Outputs)**:
- **Train-Data/EACalculation.csv**: Master preprocessed dataset containing base physical metrics and calibrated timetable slack for all 10,620 trains.
- **Attributes & Features Created / Calculated**:

Attribute Name	Data Type	Formula / Description
duration_mins	Integer	Total scheduled journey time parsed from travel_time (e.g. 40:45 Hrs. -> 2445 mins).
halt_mins	Integer	Dwell duration at each station stop parsed from halt_time (e.g. 10 Min -> 10 , 01:10 Hr -> 70).
total_halt_count	Integer	Total count of scheduled halts (count(sr_no)).
total_halt_duration	Integer	Sum of all scheduled halt dwell times across the journey (sum(halt_mins)).
total_distance_km	Float	Maximum cumulative route distance (max(dist_km)).
train_tier	String	Priority classification: T1_PREMIUM (Vande Bharat, Rajdhani, Shatabdi, Duronto), T2_SUPERFAST , or T3_EXPRESS .
effective_speed_kmph	Float	Realistic commercial cruising speed: 90.0 (T1), 68.0 (T2), 55.0 (T3).
halt_kinetic_rate	Float	Kinetic stopping and starting penalty: 1.85 mins (T1), 2.15 mins (T2), 2.35 mins (T3).
ideal_running_mins	Float	Pure cruising motion time at booked speed: (total_distance_km / effective_speed_kmph) * 60 (capped at 88% of duration).
accel_decel_delay_mins	Float	Kinetic time lost braking to and accelerating from halts: (total_halt_count - 1) * halt_kinetic_rate + halt_kinetic_rate .
gross_physical_mins	Float	Physical travel time: ideal_running_mins + total_halt_duration + accel_decel_delay_mins .
extra_time_mins	Float	Gross timetable buffer over pure cruising: max(0, duration_mins - ideal_running_mins) .
net_slack_mins	Float	Calibrated Net Slack / Cushion (EA) : Timetable margin remaining after subtracting cruising, scheduled halts, AND kinetic deceleration/acceleration: max(0, duration_mins - gross_physical_mins) .

2.2. 01bKineticAccelDecelDelay.py

- **File Name**: **BlackSheep/Pipelines/01bKineticAccelDecelDelay.py**
- **Exact Role & Purpose**:

Simulates Newtonian physical dynamics of heavy locomotive traction and braking deceleration. Uses the **Davis Drag Equation** ($\$R = A + Bv + Cv^2\$$) and locomotive tractive effort curves (WAP-7, WAP-5, WAP-4, WDP-4D) to compute the exact kinetic time loss per scheduled stop and unplanned Temporary Speed Restriction (TSR) caution order.

Key Operational Principle: It separates the nominal kinetic time already budgeted into the Indian Railways Working Time Table (WTT) from the **unplanned kinetic delay** caused by train overload drag and unscheduled TSR slow-down cycles.

- **Referenced Files (Inputs):**
- **Train-Data/EACalculation.csv:** Train tiers, halt counts, and route endpoints.
- **Created Files (Outputs):**
- **Train-Data/SlowAccDelay.csv:** Physical kinetic acceleration and deceleration profile dataset.
- **Attributes & Features Created / Calculated:**

Attribute Name	Data Type	Formula / Description
<code>governing_cruise_speed_kmph</code>	Float	$\min(\text{track_speed_limit}, \text{coach_max_speed}, \text{loco_max_speed})$.
<code>total_train_weight_tonnes</code>	Float	Rolling stock mass: locomotive tare weight + coach tare + passenger payload (tonnes).
<code>braking_deceleration_ms2</code>	Float	Emergency/service braking deceleration rate: 0.80 (T1 disc), 0.70 (T2 air), 0.60 (T3 clasp).
<code>effective_acceleration_ms2</code>	Float	Net tractive acceleration after Davis aerodynamic and rolling resistance: $(\text{TractiveEffort} - \text{DavisDrag}) / (\text{RotaryInertia} * \text{TotalMass})$.
<code>actual_loss_per_stop_mins</code>	Float	Physical time loss per full stop cycle: $(v1 / 2) * (1/d + 1/a) / 60$.
<code>wtt_budgeted_allowance_mins</code>	Float	WTT schedule budgeted allowance: 1.80 mins (T1), 2.15 mins (T2/T3).
<code>net_unplanned_delay_per_stop_mins</code>	Float	Excess kinetic stopping time beyond schedule allowance: $\max(0, \text{actual_loss} - \text{wtt_budgeted})$.
<code>gross_physical_kinetic_time_mins</code>	Float	Fleet gross kinetic time spent stopping and starting across all halts (~ 5.8 hrs on long-haul routes).
<code>unplanned_halt_excess_delay_mins</code>	Float	Total unplanned excess halt drag across the journey.
<code>tsr_accel_decel_delay_mins</code>	Float	Unscheduled kinetic deceleration from cruising speed to 30 km/h caution order and re-acceleration.
<code>total_accel_decel_delay_mins</code>	Float	Total unplanned kinetic delay: $\text{unplanned_halt_excess_delay_mins} + \text{tsr_accel_decel_delay_mins} (d_accel_decel)$.

2.3. 02GeneratePermutations.py

- **File Name:** **BlackSheep/Pipelines/02GeneratePermutations.py**
- **Exact Role & Purpose:**

Generates the **45 discrete operational scenario permutations** (\$5 \text{ Weather} \times 3 \text{ Congestion} \times 3 \text{ TSR}\$) for each train across the fleet (\$10,620 \times 45 = 477,900\$ scenario evaluations). It incorporates corridor sensitivity coefficients (e.g. Fog multiplier on Delhi-Howrah corridor = 3.0) and applies a 5% baseline timetable buffer deduction.

- **Referenced Files (Inputs):**
- **Train-Data/EACalculation.csv:** Fleet metadata, train tiers, durations, and stations.
- **Created Files (Outputs):**
- **Train-Data/PnCOutput.csv:** Master matrix of 477,900 evaluated scenario permutations.
- **Attributes & Features Created / Calculated:**

Attribute Name	Data Type	Formula / Description
<code>weather</code>	String	Scenario weather condition: Clear , Fog , Heavy_Rain , Thunderstorm , Snow .
<code>priority_congestion</code>	String	Line capacity / Overtake density: None , Low , High .
<code>tsr_level</code>	String	Speed restriction intensity: None , Minor (2 zones), Major (4 zones).
<code>d_weather</code>	Float	Weather shock penalty scaled by corridor-specific weather coefficients.
<code>d_priority</code>	Float	Overtake siding detention based on train priority tier (T1 = 0m, T2 = 15-35m, T3 = 30-75m).
<code>d_tsr_accel</code>	Float	Speed restriction traversal overhead (7.5 mins per TSR zone).
<code>predicted_delay_mins</code>	Integer	Baseline primary delay: $\max(0, d_weather + d_priority + d_tsr_accel - 0.05 * duration_mins)$.

2.4. 05CascadeBlockHeadway.py

- **File Name:** `BlackSheep/Pipelines/05CascadeBlockHeadway.py`
- **Exact Role & Purpose:**

Cascading Stage 1 (M05): Simulates block section headway queuing and signal aspect deceleration. When a leading train is delayed on a route corridor zone, trailing trains traveling in the same direction cannot enter the block until the mandatory safety headway separation (default: 8.0 mins) is restored.

Key Operational Principle: Implements the **Track Occupancy Rule**: if a train blocking the route zone is delayed by \$X\$ minutes, the trailing train directly inherits that delay plus a 3.0-minute signal aspect caution drag.

- **Referenced Files (Inputs):**
 - `Train-Data/TrainSchedules_DND.csv`: Real station halt sequences and departure times.
 - `Train-Data/EACalculation.csv`: Fleet metadata and tiers.
 - `Train-Data/PnCOutput.csv`: Initial primary delay per train for the specified scenario.
 - `Train-Data/TRETARoutes.csv`: Canonical route identifiers and zone mappings.
- **Created Files (Outputs):**
 - `Train-Data/HeadwayDelay.csv`: Headway conflict simulation results.
- **Attributes & Features Created / Calculated:**

Attribute Name	Data Type	Formula / Description
<code>treta_zone_id</code>	String	Directional station-hop block section (<code>station_code</code> -> <code>next_station</code>).
<code>actual_dep_mins</code>	Float	Scheduled departure time plus accumulated primary delay (<code>dep_mins</code> + <code>primary_delay_mins</code>).
<code>headway_delay_mins</code>	Float	Detention incurred when <code>prev_actual_dep + headway > scheduled_dep</code> : $\max(\text{prev_primary_delay}, \text{prev_actual_dep} + \text{headway} - \text{dep_mins})$.
<code>signal_aspect_delay_mins</code>	Float	Deceleration penalty (3.0 mins) incurred approaching yellow/double-yellow aspects behind an occupied block.
<code>sections_with_headway_conflict</code>	Integer	Total count of route block sections where headway queuing occurred.

Attribute Name	Data Type	Formula / Description
max_single_headway_delay	Float	Maximum single headway detention on any block section.
total_headway_delay_mins	Float	Fleet-aggregated headway queuing delay (capped at 120.0 mins).
cumulative_delay_after_m05	Float	Running total delay: <code>primary_delay_mins + total_headway_delay_mins</code> .

2.5. 06CascadeJunctionStarvation.py

- **File Name:** `BlackSheep/Pipelines/06CascadeJunctionStarvation.py`
- **Exact Role & Purpose:**

Cascading Stage 2 (M06): Simulates platform allocation bottlenecks and outer home signal detention at major railway junctions and terminal hubs (e.g. New Delhi, Howrah, Mumbai CSMT).

Key Operational Principle: Uses an event-driven **Min-Heap Priority Queue** tracking real-time platform release times. When all physical platforms at a station are occupied by delayed trains, incoming arrivals are held at outer home signals until a reception line is cleared.

- **Referenced Files (Inputs):**
 - `Train-Data/HeadwayDelay.csv`: Upstream cumulative delays from Stage 1.
 - `Train-Data/TrainSchedules_DND.csv`: Station arrival timings, dwell times, and station traffic densities.
- **Created Files (Outputs):**
 - `Train-Data/PlatformAllocDelay.csv`: Junction platform starvation simulation results.
- **Attributes & Features Created / Calculated:**

Attribute Name	Data Type	Formula / Description
platform_capacity	Integer	Dynamically assigned platform berths based on call volume (e.g. ≥ 400 calls = 12 platforms; ≥ 200 = 8; ≥ 100 = 5).
platform_release_mins	Float	Timestamp when a berthing train vacates the platform: <code>actual_arr_mins + dwell_mins + clearance_margin_mins</code> .
outer_signal_delay_mins	Float	Detention spent waiting at the home signal: <code>max(0, earliest_free_platform_time - actual_arr_mins)</code> .
junctions_starved	Integer	Number of junction stations where outer home signal detention was suffered.
max_junction_delay	Float	Maximum detention experienced at any single junction.
total_junction_delay_mins	Float	Total platform starvation delay across the trip (capped at 45.0 mins).
cumulative_delay_after_m06	Float	Running total delay: <code>cumulative_delay_after_m05 + total_junction_delay_mins</code> .

2.6. 07CascadeTurnaroundLinkage.py

- **File Name:** `BlackSheep/Pipelines/07CascadeTurnaroundLinkage.py`
- **Exact Role & Purpose:**

Cascading Stage 3 (M07): Simulates secondary propagation across shared rolling stock rakes.

Key Operational Principle: Identifies paired reverse train services (e.g. 12001 NDLS-BPL Shatabdi paired with 12002 BPL-NDLS Shatabdi). When an inbound train arrives late at its terminal, the paired outbound service cannot depart on time if the remaining layover is shorter than the mandatory pit-line maintenance, secondary examination, and cleaning buffer (60 mins for T1, 90 mins for T2, 120 mins for T3).

- **Referenced Files (Inputs):**
- **Train-Data/PlatformAllocDelay.csv:** Upstream cumulative delays from Stage 2.
- **Train-Data/EACalculation.csv:** Fleet tiers and endpoint stations.
- **Train-Data/TrainSchedules_DND.csv:** Origin departure and destination arrival timetable records.
- **Created Files (Outputs):**
- **Train-Data/ServicingDelay.csv:** Rake turnaround linkage simulation results.
- **Attributes & Features Created / Calculated:**

Attribute Name	Data Type	Formula / Description
paired_train_no	Integer	Matched paired service sharing the same physical rake ($\text{cand} = \text{t1} + 1$ if $\text{t1} \% 2 \neq 0$ else $\text{t1} - 1$).
actual_dest_arr_mins	Float	Inbound train actual arrival time at terminal: $\text{dest_arr_mins} + \text{cumulative_delay_after_m06}$.
origin_dep_mins	Float	Scheduled outbound departure time of paired service.
turnaround_delay_mins	Float	Delay propagated to outbound service: $\max(0, \text{required_buffer} - (\text{outbound_sched_dep} - \text{inbound_actual_arr}))$ (capped at 120.0 mins).
cumulative_delay_after_m07	Float	Running total delay: $\text{cumulative_delay_after_m06} + \text{turnaround_delay_mins}$.

2.7. 08CascadeCrossingConflicts.py

- **File Name:** BlackSheep/Pipelines/08CascadeCrossingConflicts.py
- **Exact Role & Purpose:**

Cascading Stage 4 (M08): Simulates bidirectional track conflicts on single-line corridors.

Key Operational Principle: Detects simultaneous opposing movements (UP vs DOWN trains) entering a single-track block between two stations. Resolves crossing precedence using **Tier Precedence Ranking** ($\text{T1_PREMIUM} > \text{T2_SUPERFAST} > \text{T3_EXPRESS}$). The subordinate train is held in a crossing loop siding while the higher-priority opposing train traverses the single line.

- **Referenced Files (Inputs):**
- **Train-Data/ServicingDelay.csv:** Upstream cumulative delays from Stage 3.
- **Train-Data/TrainSchedules_DND.csv:** Station sequence timings and directions.
- **Train-Data/EACalculation.csv:** Fleet tiers for precedence ranking.
- **Created Files (Outputs):**
- **Train-Data/SingleTrackDelay.csv:** Crossing conflict simulation results.
- **Attributes & Features Created / Calculated:**

Attribute Name	Data Type	Formula / Description
corridor	String	Undirected block segment identifier ($\text{station_A} \leftrightarrow \text{station_B}$).
direction	String	Traffic direction: UP (alphabetical order) or DOWN .

Attribute Name	Data Type	Formula / Description
rank	Integer	Precedence ranking: 3 (T1 Premium), 2 (T2 Superfast), 1 (T3 Express).
crossing_conflicts_count	Integer	Total count of crossing loop stops endured by the subordinate train.
max_single_crossing_delay	Float	Maximum single siding wait time awaiting opposing train clearance.
total_crossing_delay_mins	Float	Total crossing loop detention across the route: (Opposing Delay + Traversal Time + Clearance) (capped at 45.0 mins).
cumulative_delay_after_m08	Float	Running total delay: cumulative_delay_after_m07 + total_crossing_delay_mins .

2.8. 09CascadeCrewDutyExpiry.py

- **File Name:** **BlackSheep/Pipelines/09CascadeCrewDutyExpiry.py**
- **Exact Role & Purpose:**

Cascading Stage 5 (M09): Simulates continuous duty timeouts of running staff (Loco Pilots, Assistant Loco Pilots, and Train Managers) under statutory Indian Railways **Hours of Employment and Regulation (HOER)** rules (statutory 10-hour limit).

Key Operational Principle: When cumulative operational delays push continuous duty beyond 10 hours and cumulative delay exceeds 45 minutes, the train is halted at the nearest crew lobby awaiting mobilization of an emergency standby relief crew (45 to 75 minutes penalty).

- **Referenced Files (Inputs):**
- **Train-Data/SingleTrackDelay.csv:** Upstream cumulative delays from Stage 4.
- **Train-Data/EACalculation.csv:** Journey durations, distances, and allotted buffers.
- **Created Files (Outputs):**
- **Train-Data/TiredCrewDelay.csv:** Crew duty expiry simulation results.
- **Attributes & Features Created / Calculated:**

Attribute Name	Data Type	Formula / Description
duty_limit_exceeded	Boolean	True if (duration_mins + cumulative_delay_after_m08 > 600.0) AND cumulative_delay_after_m08 >= 45.0 .
crew_relief_delay_mins	Float	Emergency crew lobby mobilization wait time (random uniform 45.0 to 75.0 mins).
total_cascading_delay_mins	Float	Total accumulated cascading delay: d_headway + d_junction + d_turnaround + d_crossing + d_crew .
gross_delay_mins	Float	Total Gross Disturbance: primary_delay_mins + total_cascading_delay_mins .
cumulative_delay_after_m09	Float	Equal to gross_delay_mins .

2.9. CalculateExtraTime.py

- **File Name:** **BlackSheep/Pipelines/CalculateExtraTime.py**
- **Exact Role & Purpose:**

Auditing and statistical verification utility for timetable slack engineering. Compares hypothetical operational policies:

- 1. Nominal Average Speeds (85 km/h for T1, 45 km/h for T2/T3).
- 2. 90% of Maximum Permissible Speed (MPS: 117 km/h for T1, 99 km/h for T2/T3).

Key Result: Proves that nominal average speeds yield negative extra time for 100% of Superfast trains (because Indian Railways statutory Superfast minimum commercial speed is 55 km/h), confirming why the calibrated physics-based formula in [01Preprocess.py](#) is mathematically required.

- **Referenced Files (Inputs):**
- [Train-Data/EACalculation.csv](#): Fleet metadata, durations, distances, halts, and slack values.
- **Created Files (Outputs):**
- Outputs audit tables directly to terminal/logs (does not overwrite CSVs).
- **Attributes & Features Evaluated:**
- [nom_ideal](#), [nom_extra](#), [neg_pct](#), [eff_v](#), [Negative Buffer %](#).

2.10. GenerateMasterPnC.py

- **File Name:** [BlackSheep/Pipelines/GenerateMasterPnC.py](#)
- **Exact Role & Purpose:**

Constructs the comprehensive **Master Permutation & Combination Matrix ([master_pnc](#))** across **10 discrete operational disturbance dimensions** (\$3 \times 5 \times 3 \times 3 \times 4 \times 3 \times 3 \times 2 \times 2 \times 2 = 38,880 \text{ unique combinations}\$).

Computes the exact deterministic compounding delay resolution for every permutation and uploads the result directly to Neon PostgreSQL table [master_pnc](#) with a local CSV backup.

- **Referenced Files (Inputs):**
- None (Generates pure Cartesian product algorithmically).
- **Created Files (Outputs):**
- [Train-Data/masterPnC.csv](#): 38,880 rows of scenario disturbance resolutions (~6.5 MB).
- Neon PostgreSQL Table: [master_pnc](#) (with primary key on [combination_id](#)).
- **Attributes & Features Created / Calculated:**

Attribute Name	Category	Description
combination_id	Identifier	Unique surrogate key (1 to 38,880).
train_tier	Dimension	T1_PREMIUM , T2_SUPERFAST , T3_EXPRESS_PASSENGER .
weather	Dimension	Clear , Fog , Heavy_Rain , Thunderstorm , Snow .
tsr_level	Dimension	None , Minor , Major .
priority_congestion	Dimension	None , Low , High .
treta_block_occupancy	Dimension	Track_Clear , Preceding_Delayed_Minor , Preceding_Delayed_Moderate , Preceding_Delayed_Severe .
crossing_conflict	Dimension	Double_Quad_Track , Minor_Crossing_Wait , Major_Crossing_Wait .
alarm_chain_pulling	Dimension	0_Events , 1_Event , 2_Events .
engine_failure	Dimension	Nominal , Failure .
terminal_platform_hold	Dimension	Platform_Available , Outer_Holding .

Attribute Name	Category	Description
crew_duty_status	Dimension	Duty_Valid, Duty_Exceeded.
primary_delay_mins	Metric	d_weather + d_tsr + d_acp + d_engine_failure.
cascade_delay_mins	Metric	d_headway_treta + d_crossing + d_platform_hold + d_crew_delay.
gross_delay_mins	Metric	primary_delay_mins + cascade_delay_mins.
nominal_ea_buffer_mins	Metric	Nominal timetable buffer: 45m (T1), 90m (T2), 135m (T3).
absorbed_by_ea_mins	Metric	Buffer recovery: $\min(\text{gross_delay_recovery_rate}, \text{nominal_ea} \cdot 0.20)$.
net_delay_mins	Metric	$\max(0, \text{gross_delay} - \text{absorbed_by_ea})$.
arrival_status	Metric	Strict lateness classification: ON_TIME if net_delay <= 5.0 mins, else LATE.

2.11. GenerateTRETARoutes.py

- **File Name:** BlackSheep/Pipelines/GenerateTRETARoutes.py
- **Exact Role & Purpose:**

Generates the canonical **TRETAroute_number** identifier for all 10,620 trains and partitions each train's route into sequential station-hop **treta_zone_element** segments.

This establishes the common spatial coordinate system enabling the simulation of trailing headway queuing, engine failure shockwaves, and track occupancy ripple effects.

- **Referenced Files (Inputs):**
 - **Train-Data/TrainSchedules_DND.csv:** Real timetable stops and sequences.
 - **Train-Data/TrainsMetadata_DND.csv:** Source and destination endpoints.
 - **Train-Data/EACalculation.csv:** Existing preprocessed dataset (to inject **treta_route_number**).
- **Created Files (Outputs):**
 - **Train-Data/TRETARoutes.csv:** Mapping of each train to its canonical **treta_route_number** and total route zones.
 - **Train-Data/TRETAZoneElements.csv:** Detailed table of 200,126 directional station-to-station block hops.
 - Updates **Train-Data/EACalculation.csv** with column **treta_route_number**.
- **Attributes & Features Created / Calculated:**

Attribute Name	Data Type	Formula / Description
treta_zone_element	String	Block segment identifier: station_code -> next_stn (e.g. NDLS->AGC).
hop_index	Integer	Sequential index of the block hop along the train's route (1 to \$N\$).
total_treta_zones	Integer	Total number of block zones along the entire route.
treta_route_number	String	Canonical route identifier: TRETA-<SRC>-<DST>-Z<total_zones>-<train_no> (e.g. TRETA-NDLS-BPL-Z10-12001).

2.12. MLCompoundDelayModel.py

- **File Name:** `BlackSheep/Pipelines/MLCompoundDelayModel.py`
- **Exact Role & Purpose:**

The core object-oriented machine learning simulation engine (`RailwayDelayMLSimulator`). Encapsulates:

- Multi-domain scenario vector evaluation across 6 operational domains (Weather, Track, Dispatch, Traction, Terminal, Crew).
- Live simulation of abnormal failure events:
- **Alarm Chain Pulling (ACP):** Direct +15.0 mins primary shock per event.
- **Preceding Train Block on Treta Zone:** Trailing train directly inherits blocking train delay.
- **Locomotive Engine Failure:** +15.0 mins direct delay plus +12.5 mins downstream zone ripple propagation.
- Matrix generation utility (`--generate_matrix`) producing `Train-Data/MLCompoundPermutations.csv`.
- **Referenced Files (Inputs):**
- `Train-Data/EACalculation.csv`: Preprocessed fleet metadata and calibrated EA slack.
- **Created Files (Outputs):**
- `Train-Data/MLCompoundPermutations.csv` (when executed with `--generate_matrix`).
- **Attributes & Features Created / Calculated:**
- Breakdown dictionaries: `primary_breakdown` (`d_weather`, `d_track`, `d_traction`, `d_kinetic`, `d_chain_pulling`, `d_engine_failure`, `PrimaryDelay`).
- Cascading breakdown: `cascade_breakdown` (`d_headway`, `d_junction`, `d_turnaround`, `d_crossing`, `d_crew`, `CascadeDelay`).
- Mathematical resolution: `math_resolution` (`grossDelay`, `EA_allotted_mins`, `Absorbed_by_EA`, `NetDelay`, `Compound ETA`, `Arrival_Status`).

2.13. Orchestrar.py

- **File Name:** `BlackSheep/Pipelines/Orchestrar.py`
- **Exact Role & Purpose:**

The **Master Pipeline Orchestrator**. Dynamically imports and executes the entire cascading pipeline sequentially:

1. `01bKineticAccelDecelDelay` $\rightarrow$ Computes kinetic drag (`SlowAccDelay.csv`).
 2. `05CascadeBlockHeadway` $\rightarrow$ Simulates headway queuing (`HeadwayDelay.csv`).
 3. `06CascadeJunctionStarvation` $\rightarrow$ Simulates platform outer signal detention (`PlatformAllocDelay.csv`).
 4. `07CascadeTurnaroundLinkage` $\rightarrow$ Simulates rake turnaround delay (`ServicingDelay.csv`).
 5. `08CascadeCrossingConflicts` $\rightarrow$ Simulates single-line crossing loop detention (`SingleTrackDelay.csv`).
 6. `09CascadeCrewDutyExpiry` $\rightarrow$ Simulates HOER relief timeout (`TiredCrewDelay.csv`).
 7. **Consolidation:** Computes `PrimaryDelay`, `CascadeDelay`, `grossDelay`, applies tier-based recovery factors against calibrated Net Slack (`EA_allotted_mins`), and derives `NetDelay`, `Final ETA mins`, and `Arrival_Status`.
- **Referenced Files (Inputs):**
 - `Train-Data/EACalculation.csv`, `Train-Data/TrainSchedules_DND.csv`, `Train-Data/PnCOutput.csv`, `Train-Data/TRETARoutes.csv`.
 - Upstream intermediate files generated during the pipeline run.
 - **Created Files (Outputs):**

- **Train-Data/FinalOrchestraOutput.csv**: The master consolidated fleet prediction dataset for all 10,620 trains.
- **Attributes & Features Created / Calculated**:

Attribute Name	Formula / Description
d_weather_tsr_priority	Upstream primary shock delay from weather, speed restrictions, and overtake sidings.
d_accel_decel	Unplanned kinetic deceleration & acceleration loss from SlowAccDelay.csv .
PrimaryDelay	Total primary disturbance: d_weather_tsr_priority + d_accel_decel .
d_headway	Block section headway queuing delay (Stage 1).
d_junction	Outer home signal detention from occupied junction platforms (Stage 2).
d_turnaround	Delayed departure caused by late inbound paired rake turnaround (Stage 3).
d_crossing	Single-track crossing loop detention awaiting opposing higher-priority train (Stage 4).
d_crew	Emergency relief crew standby waiting penalty when 10-hr HOER duty is exceeded (Stage 5).
CascadeDelay	d_headway + d_junction + d_turnaround + d_crossing + d_crew .
grossDelay	Total physical disturbance: PrimaryDelay + CascadeDelay .
EA_allotted_mins	Calibrated Net Slack buffer from EACalculation.csv .
Absorbed_by_EA_mins	Buffer recovery: min(grossDelay recovery_rate, EA_allotted_mins 0.50) where recovery rate = 0.40 (T1), 0.25 (T2), 0.15 (T3).
NetDelay	Net destination arrival delay: max(0.0, grossDelay - Absorbed_by_EA_mins) .
ScheduledArrival_mins	Timetable scheduled arrival timestamp (equal to journey duration_mins).
Final_ETA_mins	Predicted actual arrival timestamp: ScheduledArrival_mins + NetDelay .
Arrival_Status	Punctuality verdict: ON_TIME if NetDelay <= 5.0 mins, else LATE .

2.14. __init__.py

- **File Name**: **BlackSheep/Pipelines/__init__.py**
- **Exact Role & Purpose**:

Python package initialization marker. Allows modules within **Pipelines/** to be cleanly imported across **AppServer.py**, **SheepBand.py**, and root scripts as a native Python namespace package.

- **Referenced Files**: None.
- **Created Files**: None.
- **Attributes Created**: None.

3. End-to-End File Dependency & Artifact Matrix

Script Name	Input Files Referenced	Output Files Generated / Updated	Core Output Attributes
01Preprocess.py	TrainsMetadata_DND.csv TrainSchedules_DND.csv	Train-Data/EACalculation.csv	duration_mins, total_halt_count, total_halt_duration, ideal_running_mins, extra_time_mins, net_slack_mins, train_tier
01bKineticAccelDecelDelay.py	EACalculation.csv	Train-Data/SlowAccDelay.csv	governing_cruise_speed_kmph, total_train_weight_tonnes, actual_loss_per_stop_mins, total_accel_decel_delay_mins
02GeneratePermutations.py	EACalculation.csv	Train-Data/PnCOutput.csv	weather, priority_congestion, tsr_level, predicted_delay_mins (45 permutations/train)
05CascadeBlockHeadway.py	TrainSchedules_DND.csv EACalculation.csv PnCOutput.csv TRETARoutes.csv	Train-Data/HeadwayDelay.csv	treta_zone_id, actual_dep_mins, headway_delay_mins, total_headway_delay_mins, cumulative_delay_after_m05
06CascadeJunctionStation.py	HeadwayDelay.csv TrainSchedules_DND.csv	Train-Data/PlatformAllocDelay.csv	platform_capacity, outer_signal_delay_mins, total_junction_delay_mins, cumulative_delay_after_m06
07CascadeTurnaroundLinkage.py	PlatformAllocDelay.csv EACalculation.csv TrainSchedules_DND.csv	Train-Data/ServicingDelay.csv	paired_train_no, turnaround_delay_mins, cumulative_delay_after_m07
08CascadeCrossingConflicts.py	ServicingDelay.csv TrainSchedules_DND.csv EACalculation.csv	Train-Data/SingleTrackDelay.csv	corridor, crossing_conflicts_count, total_crossing_delay_mins, cumulative_delay_after_m08
09CascadeCrewDutyExpiry.py	SingleTrackDelay.csv EACalculation.csv	Train-Data/TiredCrewDelay.csv	duty_limit_exceeded, crew_relief_delay_mins, total_cascading_delay_mins, gross_delay_mins
CalculateExtraTime.py	EACalculation.csv	None (Audit Report)	nom_ideal, nom_extra, Negative Buffer % evaluation
GenerateMasterPnC.py	None (Combinatorial generator)	Train-Data/masterPnC.csv Neon DB Table: master_pnc	combination_id, 10 condition dimensions, primary_delay_mins, cascade_delay_mins, gross_delay_mins, net_delay_mins, arrival_status (38,880 rows)
GenerateTRETARoutes.py	TrainSchedules_DND.csv TrainsMetadata_DND.csv EACalculation.csv	Train-Data/TRETARoutes.csv Train-Data/TRETAZoneElements.csv Updates EACalculation.csv	treta_route_number, treta_zone_element, hop_index, total_treta_zones
MLCompoundDelayModel.py	EACalculation.csv	Train-Data/MLCompoundPermutations.csv (Optional)	Multi-domain delay vectors, ACP and engine failure breakdown, grossDelay, Absorbed_by_EA, NetDelay, Arrival_Status
Orchestrar.py	All upstream pipeline artifacts & intermediate CSVs	Train-Data/FinalOrchestraOutput.csv	Master synthesis: PrimaryDelay, CascadeDelay, grossDelay, EA_allotted_mins, Absorbed_by_EA_mins, NetDelay, Final_ETA_mins, Arrival_Status

4. Mathematical Compounding Delay Resolution Summary

The cascading pipeline resolves every train journey through six governing mathematical formulations:

```
[PRIMARY DISTURBANCES]
D_Primary = d_weather + d_TSR + d_overtake + d_accel_decel + d_ACP + d_engine

[SECONDARY CASCADING PROPAGATION]
D_Cascade = d_headway + d_junction + d_turnaround + d_crossing + d_crew

[TOTAL GROSS DISTURBANCE]
D_Gross = D_Primary + D_Cascade

[TIMETABLE SLACK (EA) RECOVERY]
D_Absorbed = min(D_Gross, R_Tier * EA_allotted)
where R_Tier:
    • T1_PREMIUM      = 0.40  (40% dynamic buffer recovery)
    • T2_SUPERFAST    = 0.25  (25% dynamic buffer recovery)
    • T3_EXPRESS      = 0.15  (15% dynamic buffer recovery)

[FINAL NET DESTINATION DELAY]
D_Net = max(0.0, D_Gross - D_Absorbed)

[ESTIMATED TIME OF ARRIVAL & STATUS]
Final_ETA = Scheduled_Arrival + D_Net
Arrival_Status = "ON_TIME" if D_Net <= 5.0 mins else "LATE"
```

[NOTE] Summary of Key Takeaways: 1. All physical, speed restriction, and kinetic loss delays feed into **PrimaryDelay**. 2. All corridor headway, outer signal holding, paired rake servicing, single-track crossing, and crew timeout delays feed into **CascadeDelay**. 3. Timetable slack (**EA**) dynamically absorbs disturbances based on dispatch priority tiers. 4. The strict lateness threshold across Indian Railways simulation is **5.0 minutes**.